Date:

EXPERIMENT NO. 7

AIM: Traffic Data Analysis using Wireshark.

Tools required:

- Desktop computer
- Wireshark

Submission: Written file submission (starting from page no. 4) is expected.

References
Wireshark Installation : Wireshark Tutorial for Beginners - Installation
Wireshark Walkthrough : https://www.youtube.com/watch?v=lb1Dw0elw0Q

Wireshark Videos:

- Edit Windows
- Canvas Window
- Capturing Data
- Interpreting data in Wireshark
- Ethernet Frame
- IP Frame
- Throughput
- Flow diagram

Understanding Require

- IP Address
- URL
- Finding IP address from URL
- Finding location from IP Address

Wireshark for Windows

Install wireshark from below link

https://www.wireshark.org/#download

Wireshark for Linux
Installing Wireshark on Linux can be a little different depending on the Linux distribution.
From a terminal prompt, run these commands:
sudo apt-get install wireshark
sudo dpkg-reconfigure wireshark-common
sudo adduser $USER wireshark

Those commands download the package, update the package, and add user privileges to run Wireshark.

Capturing Data Packets on Wireshark

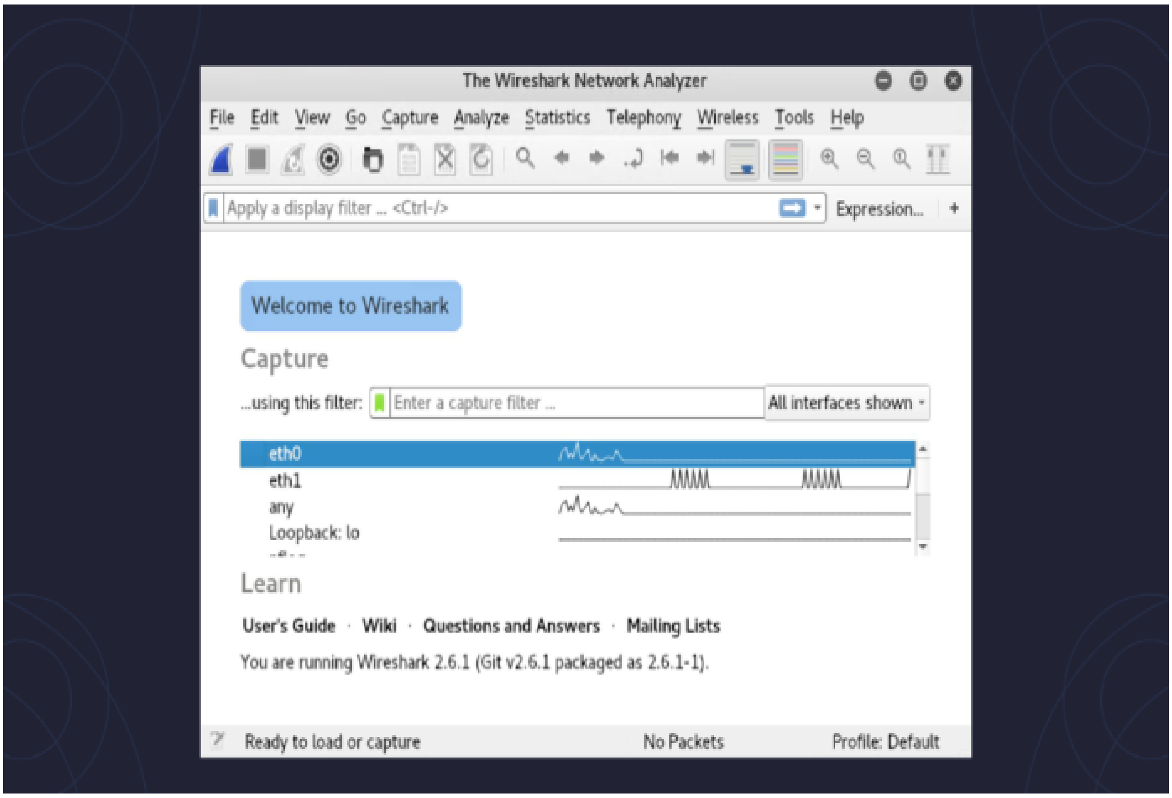


Figure - 7.1

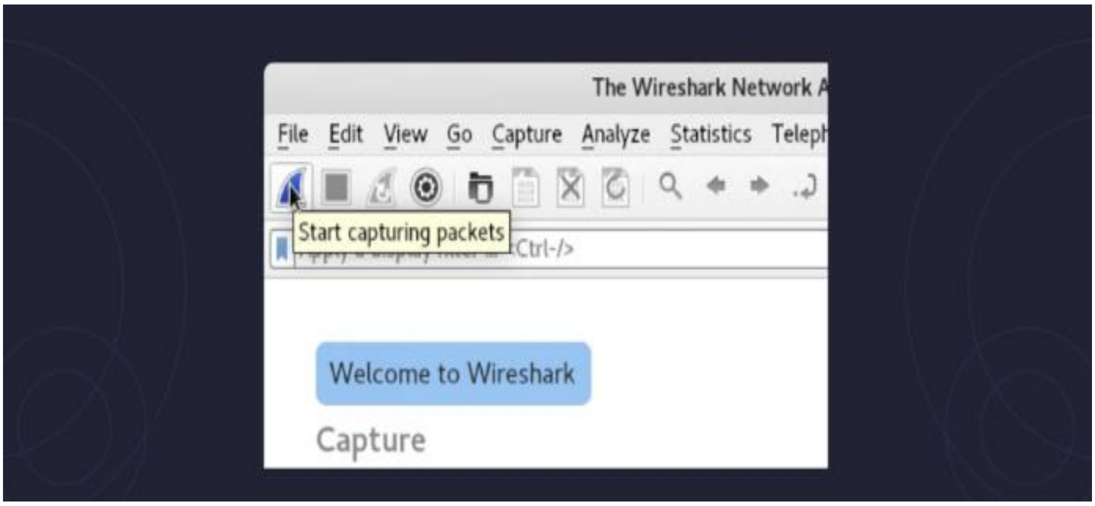


Figure -7.2

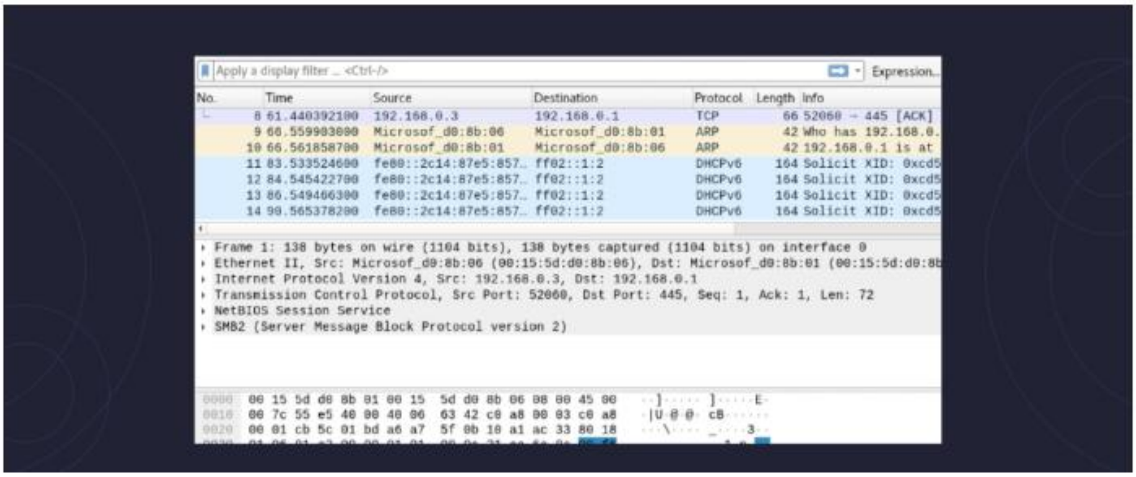


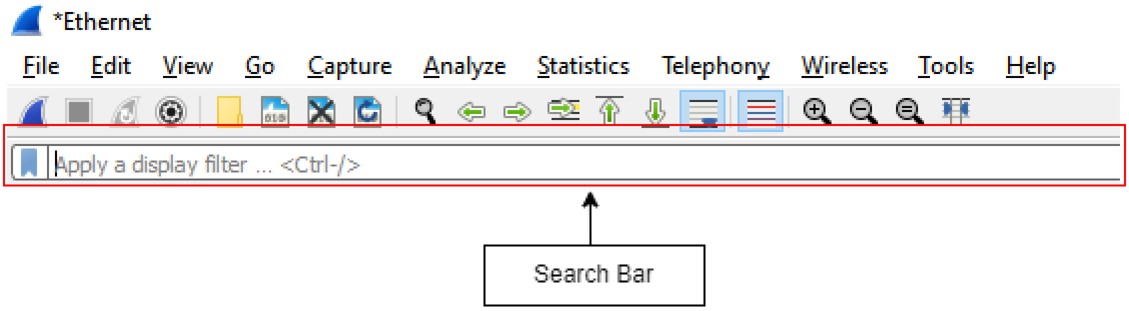
Figure - 7.3

Exercise

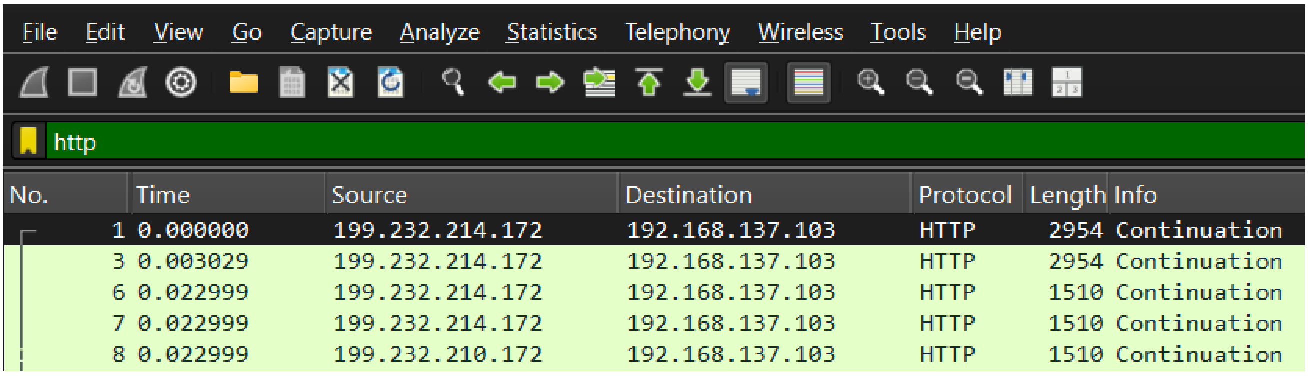
1. Turn on wireshark.
2. Then select your respective network adapter (eg. Ethernet) and start( ) capturing the data.
3. Now, go to your browser.
4. Type 172.16.12.25 in your search engine.
5. After 2 seconds go to wireshark and stop( ) capturing data.
6. Go to the wireshark file menu and click on save.
7. Save your file as pr-7_xyz_sample1.
8. Repeat steps 1,2,3.
9. Type 172.16.0.1:8090 in your search engine.
10. After 2 seconds go to wireshark and stop( ) capturing data.
11. Go to the wireshark file menu and click on save.
12. Save your file as pr-7_xyz_sample2.

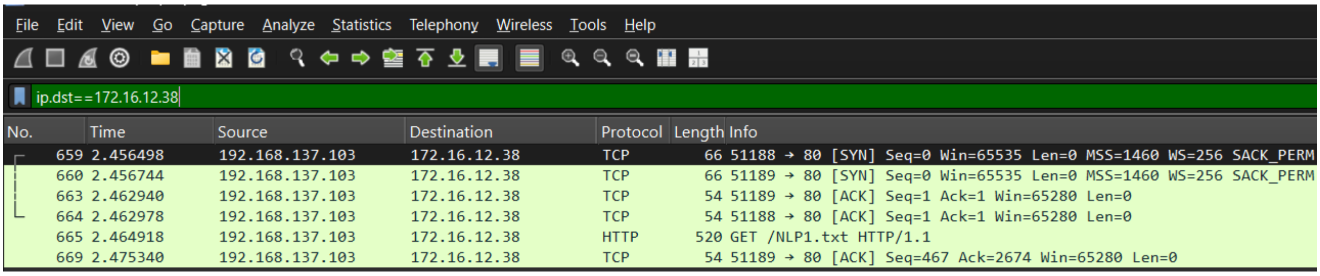
(where xyz is last three digits of your enrolment number.)

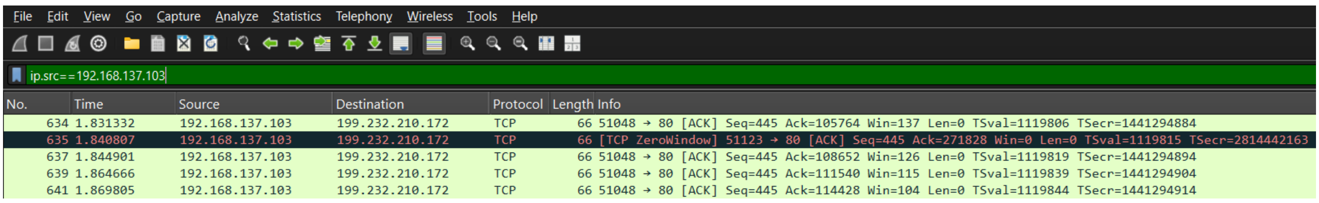
Based on this above steps, Insert screenshots of sample 1:

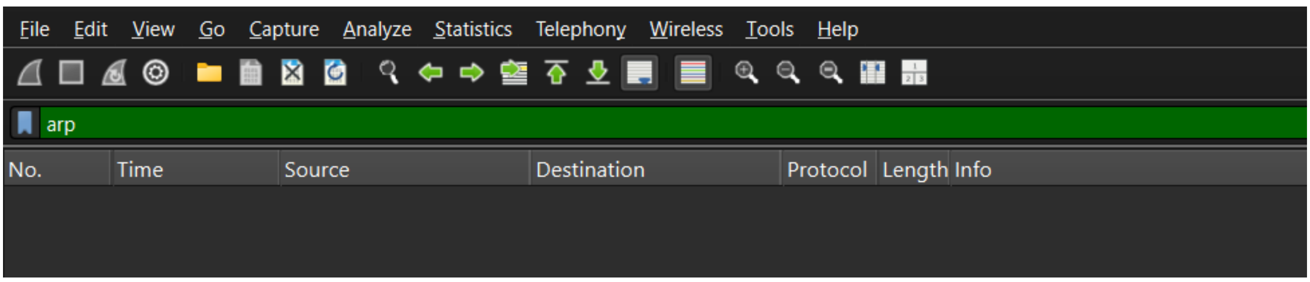


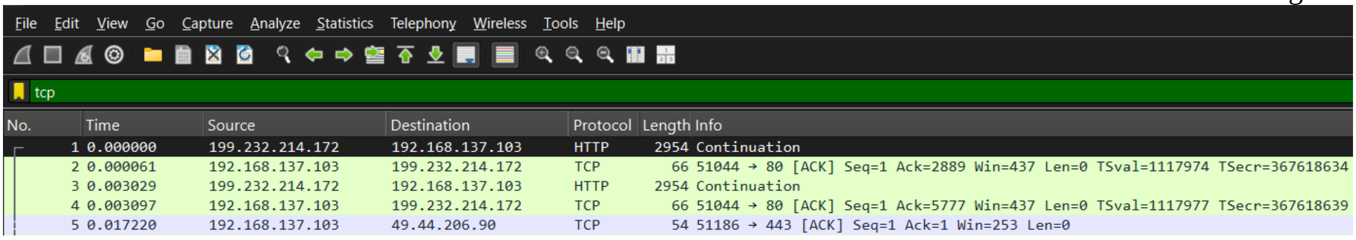
Use Different Filters and insert screenshots for sample 1.







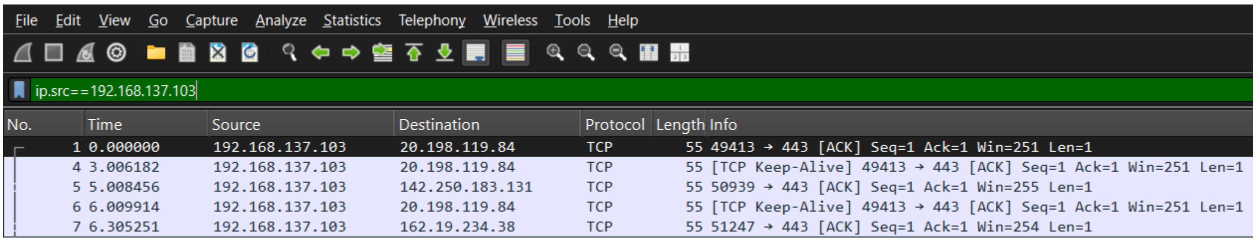




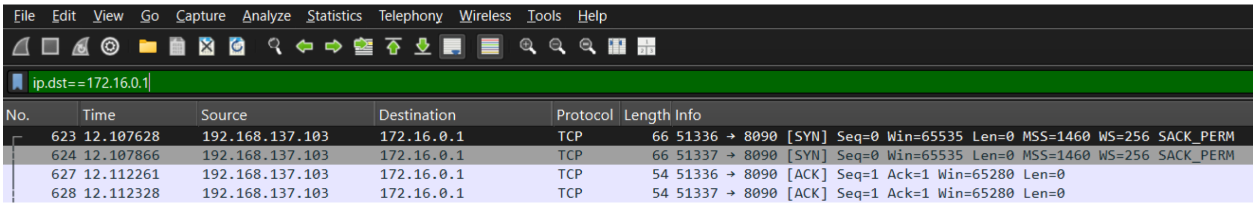
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	199.232.214.172	192.168.137.103	HTTP	2954	Continuation
2	0.000061	192.168.137.103	199.232.214.172	TCP	66	51044 → 80 [ACK] Seq=1 Ack=2889 Win=437 Len=0 TSval=1117974 TSecr=367618634
3	0.003029	199.232.214.172	192.168.137.103	HTTP	2954	Continuation
4	0.003097	192.168.137.103	199.232.214.172	TCP	66	51044 → 80 [ACK] Seq=1 Ack=5777 Win=437 Len=0 TSval=1117977 TSecr=367618639
5	0.017220	192.168.137.103	49.44.206.90	TCP	54	51186 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=0

Observe the capture data and insert screenshots for sample 2.

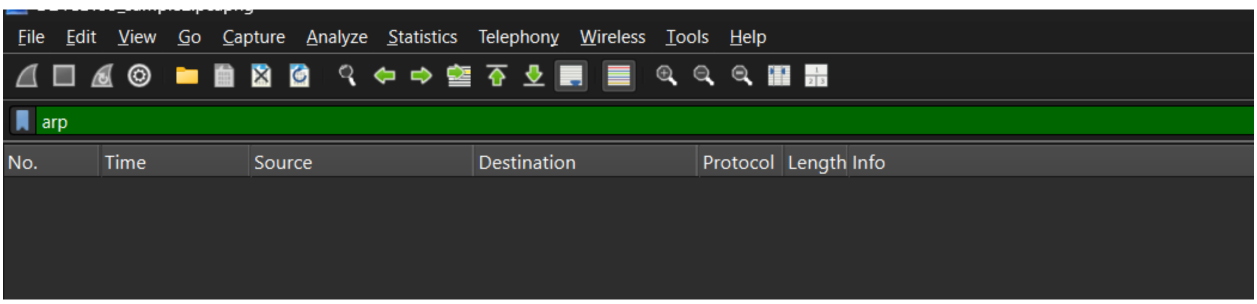
Use Different Filters and insert screenshots for sample 2.

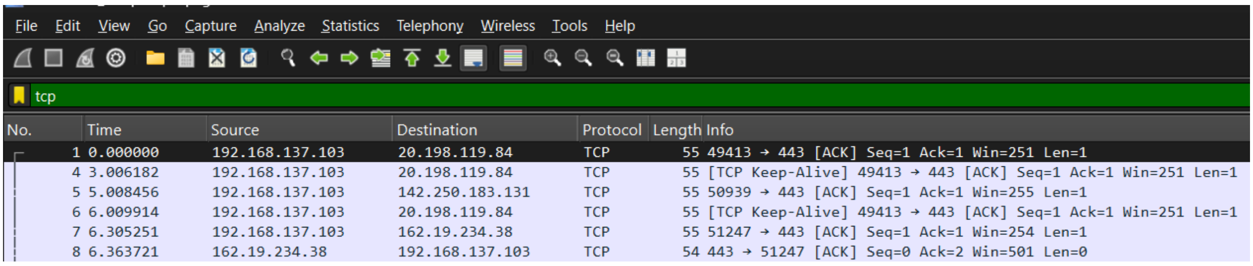


No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.137.103	20.198.119.84	TCP	55	49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
4	3.006182	192.168.137.103	20.198.119.84	TCP	55	[TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
5	5.008456	192.168.137.103	142.250.183.131	TCP	55	50939 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
6	6.009914	192.168.137.103	20.198.119.84	TCP	55	[TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
7	6.305251	192.168.137.103	162.19.234.38	TCP	55	51247 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1

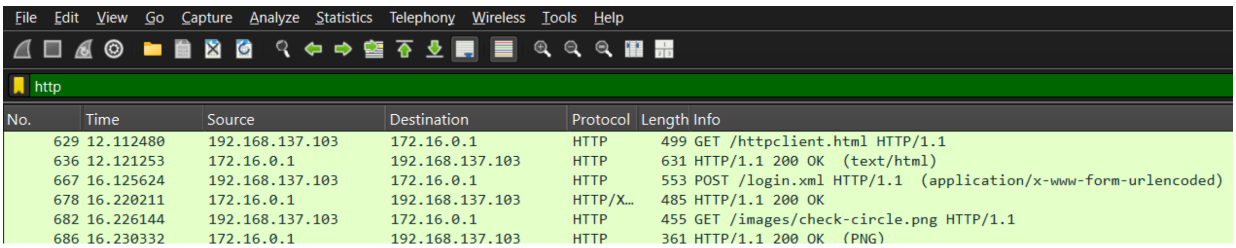


No.	Time	Source	Destination	Protocol	Length	Info
623	12.107628	192.168.137.103	172.16.0.1	TCP	66	51336 → 8090 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
624	12.107866	192.168.137.103	172.16.0.1	TCP	66	51337 → 8090 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
627	12.112261	192.168.137.103	172.16.0.1	TCP	54	51336 → 8090 [ACK] Seq=1 Ack=1 Win=65280 Len=0
628	12.112328	192.168.137.103	172.16.0.1	TCP	54	51337 → 8090 [ACK] Seq=1 Ack=1 Win=65280 Len=0





No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.137.103	20.198.119.84	TCP	55	49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
4	3.006182	192.168.137.103	20.198.119.84	TCP	55	[TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
5	5.008456	192.168.137.103	142.250.183.131	TCP	55	50939 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
6	6.009914	192.168.137.103	20.198.119.84	TCP	55	[TCP Keep-Alive] 49413 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1
7	6.305251	192.168.137.103	162.19.234.38	TCP	55	51247 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
8	6.363721	162.19.234.38	192.168.137.103	TCP	54	443 → 51247 [ACK] Seq=0 Ack=2 Win=501 Len=0



No.	Time	Source	Destination	Protocol	Length	Info
629	12.112480	192.168.137.103	172.16.0.1	HTTP	499	GET /httpclient.html HTTP/1.1
636	12.121253	172.16.0.1	192.168.137.103	HTTP	631	HTTP/1.1 200 OK (text/html)
667	16.125624	192.168.137.103	172.16.0.1	HTTP	553	POST /login.xml HTTP/1.1 (application/x-www-form-urlencoded)
678	16.220211	172.16.0.1	192.168.137.103	HTTP/X…	485	HTTP/1.1 200 OK
682	16.226144	192.168.137.103	172.16.0.1	HTTP	455	GET /images/check-circle.png HTTP/1.1
686	16.230332	172.16.0.1	192.168.137.103	HTTP	361	HTTP/1.1 200 OK (PNG)